

ABSTRACT

Urban traffic congestion poses significant challenges to road safety, energy efficiency, and emergency response times, especially under dynamic conditions such as varying weather, pedestrian density, and unexpected traffic surges. This paper proposes an intelligent urban traffic management framework that integrates Reinforcement Learning (RL) with Computer Vision (CV) to enable adaptive, context-aware traffic signal control. The proposed system prioritizes emergency vehicles through real-time visual detection and dynamic signal preemption, while simultaneously ensuring pedestrian safety by analyzing crosswalk occupancy and movement patterns. Environmental factors such as weather conditions are incorporated into the RL reward mechanism to enhance robustness under adverse scenarios. Furthermore, the system integrates real-time congestion data from Maps to provide a macroscopic traffic view, enabling proactive signal adjustments based on upstream and downstream traffic flow. Energy efficiency is achieved by minimizing idle times, reducing unnecessary signal switching, and optimizing green-phase durations. Experimental evaluations demonstrate improved traffic throughput, reduced emergency vehicle delay, enhanced pedestrian safety, and lower energy consumption compared to traditional fixed-time and adaptive traffic control systems. The proposed framework offers a scalable and real-world deployable solution for next-generation smart cities.

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